Nonlinear coupling of crystal-field excitations and spin waves in a rare-earth orthoferrite

Working draft[1]

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Rare-earth magnets host two species of elementary excitation—the collective spin waves of the transition-metal sublattice and the single-ion crystal-field (CEF) transitions of the rare earth—and their interaction underlies spin-reorientation transitions, magnetisation anomalies, and ultrafast optical control. Yet that interaction has essentially only ever been characterised thermodynamically, as a free-energy surface. Here we determine its operator content. For a non-Kramers ion such as Tm^{3+} in $TmFeO_3$, time reversal blocks the iron exchange field from coupling to the CEF coherences at first order: the interaction is intrinsically *nonlinear*, beginning at second order in the environment, and symmetry admits exactly two objects—an *energy face*, the exchange-modulated crystal field $W\,SS\,\lambda^1$, which converts magnon and CEF coherences into one another but never radiates, and a *moment face*, the order-induced electric dipole $g_d(\langle F_x\rangle+\delta F_x)\lambda^2$, which couples the CEF transition to light and, through its fluctuating part, radiates internal magnon–CEF sum- and difference-frequency tones. Two-dimensional terahertz spectroscopy resolves this structure term by term, because each cross peak certifies one conversion pathway. One Hamiltonian built from the two faces, the bare subsystems, one on-site hyperpolarisability and one self-absorption filter reproduces the polarisation-resolved 2D atlas from measured inputs; every assignment is verified by switching off individual terms in the simulation, and the structural absences of the atlas act as measurements—of a fully dark $1\to3$ dipole, of the iron emission weight, and of the remaining quadratic moments. Because the energy face is the same coupling surface whose thermal average drives spin reorientation, the atlas constitutes a coherent, mode-resolved measurement of that mechanism, and the entire dressed sector is predicted to collapse together at the transition while the bare sector survives.

I. INTRODUCTION

Magnetic insulators containing both transition-metal and rare-earth ions are governed by a conversation between two very different subsystems. The transition-metal sublattice orders and carries collective spin waves—SU(2) precessional dynamics, delocalised and sharp. The rare-earth ion contributes localised crystal-field (CEF) transitions—an atomic few-level ladder, in the language of an SU(N) generalised spin. In the rare-earth orthoferrites RFeO$_3$ this conversation is not a spectroscopic detail but the engine of the family's signature phenomena: the temperature-dependent occupation of the R levels rotates the Fe anisotropy and drives the spin-reorientation transition [3–5], and intense terahertz pulses steer the Fe order through the same rare-earth channel on picosecond timescales [7, 8]. Yet in nearly all of this physics the R–Fe interaction appears only through its *thermal average*—an anisotropy surface, a free energy. Which operators actually couple a crystal-field coherence to a magnon, with what symmetry, and at what order, has remained an open question, in $TmFeO_3$ and in the family at large.

Linear spectroscopy cannot answer it. Absorption measures pole positions and dipole weights, and any coupling that merely hybridises modes is absorbed into the mode basis. What linear response can never certify is a *conversion*: a coherence prepared on one transition and re-emitted on another. That is precisely what a cross peak in two-dimensional coherent spectroscopy (2DCS) measures [1, 2], and a conversion requires a specific anharmonic vertex. A cross peak at (ω_τ, ω_t) is therefore a measurement of one coupling term: the delay axis names the stored coherence, the detection axis the emitter, and the signed side constrains the pathway order. If symmetry makes the space of candidate vertices finite—and in a crystal it does—a polarisation-resolved 2D atlas can be read as the interaction Hamiltonian itself.

$TmFeO_3$ turns out to be the ideal system for this program, for a reason deeper than convenient level spacing. A single-cycle THz pulse spans all five modes of both subsystems—the quasi-ferromagnetic and quasi-antiferromagnetic magnons at $q_{\mathrm{FM}} = 0.38$ and $q_{\mathrm{AFM}} = 0.90$ THz of the Γ_2-ordered (F_x, C_y, G_z) Fe sublattice, and the three-singlet Tm^{3+} qutrit with $E_{12} = 0.50$, $E_{23} = 0.70$, $E_{13} = 1.20$ THz. But the decisive property is that the Tm levels are *non-Kramers singlets*. As we show next, this makes the magnon–CEF interaction intrinsically nonlinear—forbidden at first order, and organised at second order into exactly two symmetry-allowed objects. That structure is what a 2D atlas can confirm or destroy, peak by peak.

II. THE INTERACTION ALGEBRA OF A SINGLET ION

Write the Tm qutrit in Gell-Mann coordinates $\lambda^a = \langle\Lambda^a\rangle$, $a = 1\ldots8$: three coherences (each with a real and an imaginary quadrature), two populations. The question is which products of Fe-spin operators $\boldsymbol{S}$, light fields $\boldsymbol{E}, \boldsymbol{B}$, and λ^a can appear in the Hamiltonian and in the radiating moments. For a non-Kramers singlet ion the answer is strongly constrained by a single theorem.

Time reversal. Nondegenerate eigenstates of a real crystal-field Hamiltonian may be chosen real; in that basis every matrix element of the angular momentum $\boldsymbol{J}$ is purely imaginary and antisymmetric. Projected onto the qutrit, $\boldsymbol{J}$ has support *only* on the three time-odd quadratures $\lambda^{2,5,7}$, and the ion has rigorously no permanent moment. The iron exchange field $\boldsymbol{h}_{\rm ex}(\boldsymbol{S})$, acting linearly on the ion, therefore produces only the bilinear $K^-\,\boldsymbol{S}\cdot(P\boldsymbol{J}P)$, confined to $\lambda^{2,5,7}$—and this coupling fails on three independent grounds. Dynamically, a term linear in both $\boldsymbol{S}$ and λ hybridises the modes but cannot convert a stored coherence (no cross peak); structurally, its channels cancel over the four-sublattice orbit in the ordered state to machine precision; and statically, its one surviving component reorients the ground state. Its spectroscopic fingerprint is a hybridisation shift of the CEF lines, and the measured 2D transfer rows sit at the 1D absorption frequencies *exactly*—the mixer class is excluded at every coupling strength once the bare splitting is recalibrated to hold the dressed line fixed. The conclusion is a protection statement: *a singlet rare earth is invisible to its magnetic environment at first order.* The same property that makes non-Kramers ions robust against magnetic noise makes their interaction with a magnet begin at second order—intrinsically nonlinear.

The two faces of second-order exchange. At second order the environment enters as a pair, and the algebra of the qutrit fixes where each pair can land. The $1\leftrightarrow 2$ coherence has a $\mathcal{T}$-even quadrature λ^1 and a $\mathcal{T}$-odd quadrature λ^2, and each admits exactly one two-leg dressing:

$$H_W = \boldsymbol{S}^T W\,\boldsymbol{S}\,\lambda^1, \qquad H_g = -\,g_d\,E_z\,(F_x\,\lambda^2). \tag{1}$$

The spin pair SS is $\mathcal{T}$-even and lands on λ^1: this is the *energy face*, the exchange modulation of the crystal field, $W\sim\partial^2 E_{\rm CEF}/\partial S\,\partial S$. In Γ_2 symmetry leaves two components, $S_y^2\lambda^1$ and $S_xS_z\lambda^1$, carried at a single scale W: one tensor, whose components share the conversion tasks. Having no field leg, the energy face never radiates—it is the pure converter, and every cross-subsystem transfer must flow through it. The light–spin pair E_zS is the *moment face*: an electric field cannot reach the $\mathcal{T}$-odd λ^2 on its own, but paired with the $\mathcal{T}$-odd, mirror-breaking order parameter $\langle F_x\rangle$ it can—H_g is an order-induced electric dipole of the $1\leftrightarrow 2$ transition, $d_z^{\rm eff}=g_d(\langle F_x\rangle+\delta F_x)\lambda^2$. One constant serves three ways: the condensed part *drives* (an $E\parallel c$ field writes the E_{12} coherence) and *radiates linearly* (the magnon-to-CEF readout), while the fluctuating part $g_d\,\delta F_x\lambda^2$ is a bilinear emitter, radiating *internal sum- and difference-frequency tones* at $q_{\rm AFM}\pm E_{12}$—magnon–CEF SFG and DFG, in which the mixing element is a bilinear term of the material's transition moment rather than a crystal-optical $\chi^{(2)}$ acting on light. These sidebands are not optional: if a transition moment exists only because an order parameter condensed, the same coupling constant necessarily radiates the order parameter's fluctuations, with amplitudes locked to the condensed line.

Two further objects complete the taxonomy, both bounded rather than active. The field-assisted family $B\,S\,\lambda$ passes every symmetry test but is *exactly* inert in the polarisation geometries used here—its gating field component is never driven—and the population channel $SS\lambda^3$ feeds only incoherent observables (a rectified line and a slow buildup), leaving every coherent feature untouched. What remains is bare structure: the Fe magnon pair coupling (anisotropy plus Dzyaloshinskii–Moriya, entirely within the Fe sector), the bare driven qutrit with its dipoles, and—because the Tm $4c$ site lacks inversion while Fe at $4b$ does not—the on-site quadratic moment $m_z=\mu\lambda^2+\beta\lambda^1\lambda^2$, the SHG member of the same quadratic-moment family whose SFG/DFG members are supplied by $\delta F_x\lambda^2$.

The physical identity of the energy face deserves emphasis before any data. $\partial^2E_{\rm CEF}/\partial S\,\partial S$ is precisely the coupling surface whose thermal average—through the temperature-dependent rare-earth level occupations—rotates the Fe anisotropy and drives the spin-reorientation transition [3, 8]. Equation (1) is that surface promoted from thermodynamics to operators. A 2D atlas that resolves H_W term by term is therefore a coherent, mode-resolved, real-time measurement of the reorientation mechanism itself—and the structure makes a collective prediction: every feature carrying a factor of the order parameter (all of H_g's children, and the transfers whose static vertex contracts G_z) must collapse *together* at the transition, while the bare-structure features survive it.

III. FROM HAMILTONIAN TO ATLAS

The calculation keeps both subsystems at their natural level of description. The Fe^{3+} ($S=5/2$) sublattice is classical, with the calibrated orthoferrite Hamiltonian (exchange, single-ion anisotropy, Dzyaloshinskii–Moriya) reproducing $q_{\rm FM}$, $q_{\rm AFM}$ and Γ_2. Each Tm^{3+} is truncated to its three lowest singlets and carried as the vector λ^a. Because every single-ion term is linear in the Λ^a, the classical $\vec{\lambda}$ equation of motion *is* the exact von Neumann equation of the driven, damped qutrit, and a Boltzmann ensemble over level seeds is the exact thermal density matrix [SM, Sec. S1]: the $\mathrm{SU}(3)\otimes\mathrm{SU}(2)$ dynamics is exact at the single-ion level, and approximate only in treating the spin bath classically. The Fe–Tm sector is built by projecting $\boldsymbol{S}^TA\,\boldsymbol{J}$ into the qutrit in a transported gauge over the four Tm sublattices, and the taxonomy of Sec. II is exhaustive over that space [SM, Secs. S2–S4]. All spectra use the measured pump (spectrum peaked at 0.50 THz, negligible DC), measured linewidths as damping rates, and measured oscillator strengths as emission weights; peaks are identified blindly as 2D local maxima with prominence ≥ 1.5. A decisive advantage of this construction is that every term of Eq. (1)—every drive leg, vertex, detection term and filter—can be switched off individually; each assignment below is verified this way, and the verification ratios are quoted with the peaks.

Polarisation blocks. The Tm $4c$ site carries the mirror m ($z \to -z$), under which $d_{x,y}$ are even and d_z odd, while m_z is even and $m_{x,y}$ odd. Combined with the time-reversal statement this fixes the entire optical interface: the projected moment matrix has λ^2 carrying only μ_z and $\lambda^{5,7}$ only $\mu_{x,y}$, forcing $|1\rangle, |2\rangle$ to share a mirror parity and $|3\rangle$ to be opposite:

transition	magnetic	electric	owned by
$1 \leftrightarrow 2$	m_z (λ^2)	d_x (λ^1)	$(E \parallel a, H \parallel c)$
$1 \leftrightarrow 3$	m_x (λ^5)	d_z (λ^4)	$(E \parallel c, H \parallel a)$
$2 \leftrightarrow 3$	m_x (λ^7)	d_z (λ^6)	$(E \parallel c, H \parallel a)$

Each linear polarisation owns one block of the qutrit. An analyser selects a polarisation *state*, hence both the magnetic and the electric dipole radiating into it: there are only *two* detection operators in the atlas (Table I), the two channels sharing one are constrained to share it exactly, and the same operators serve as drive vertices—the construction is reciprocal at every level the data can test. Cross-polarised detection therefore reads exactly the block the pulse did not drive: this is the structural reason cross-polarisation isolates conversion, with two exact internal checks (the geometry-B subalgebra closure $\lambda^{4\text{–}7} \equiv 0$, verified to machine precision, and $F_z \equiv 0$ in geometry A, so that anything seen there at a magnon frequency must be Tm). Permanent (population) moments do not radiate at transition frequencies and are excluded from detection; a measured rectified line at $\omega_t \to 0$ in an $(E \parallel a, H \parallel c)$ channel would falsify this exclusion.

Because λ^1 and λ^2 are quadratures of one coherence, their magnitude spectra coincide to 10%: in *detection* the atlas is insensitive to the $E1/M1$ balance. It is the *drive* that fixes it: the electrically written coherence feeds the geometry-B transfer linearly in w_{E1}, and the measured ratio returns $w_{E1} = 0.54$ (in units of μ). Three cross-family scales are calibrated once and used consistently on both sides: w_{E1}; g_d (one parameter with a drive avatar, a linear-emission avatar, and a condensation-ratio correction $s = 0.42$ on its bilinear piece); and the Fe:Tm emission efficiency c_{Fe}, bounded from above by a null of the data (below). The last ingredient is propagation: with oscillator strength $f = 8.7$ the sample is optically thick at the E_{13} line (depth $\simeq 6$) and mildly at E_{12} (depth $\simeq 1$), and the same transmission filters the incoming field and the emitted $E1$ light—magnetic-dipole radiation passes freely. One filter, applied identically both ways, itself reciprocal.

TABLE I. The two detection operators. $w_{E1} = 0.54$ is the $E1/M1$ balance of the $1 \leftrightarrow 2$ transition, fixed by the geometry-B transfer (see text); the $1 \leftrightarrow 3$ transition is fully dark ($\mu_{13} = 0$ in both polarisations, $d_z(1 \to 3) \le 0.006$); no population generator appears.

$(E \parallel a, H \parallel c)$:	$F_z + \mu\lambda^2 + w_{E1}\,\lambda^1 + \beta\lambda^1\lambda^2$
A cross	$F_z \equiv 0$: Tm only. $\lambda^{1,2}$ resonant at E_{12}
B same	F_z alive but 50× smaller than the Tm terms
$(E \parallel c, H \parallel a)$:	$F_x + \mu_x\lambda^7 + w'_{E1}\lambda^6 + g_d(\langle F_x\rangle + \delta F_x)\lambda^2$
A same	CEF λ^6; d_z^{eff} (condensed + fluctuating); F_x bounded
B cross	F_x; Tm part $\equiv 0$ by closure

IV. THE ATLAS REALISES THE ALGEBRA

Figure 1 shows the four channels; we now read them against Sec. II. Throughout, the delay axis carries a coherence that *never radiates*—it need only be excitable and able to store phase—while the detection axis is constrained by the emitting multipole. That asymmetry does most of the work.

A. The energy face, read forward

Geometry A, cross-polarised: $(q_{\text{AFM}}, E_{12}) = 1.00$. The $H \parallel a$ Zeeman field launches the q_{AFM} magnon, which precesses at 0.90 THz through the delay. It cannot radiate here—it has no c-axis moment—but it need not: it modulates the Tm crystal field through $WS_y^2\lambda^1$. Each pulse supplies one magnon leg, so the nonlinear subtraction $M_{\text{NL}} = M_{AB} - M_A - M_B$ keeps the cross term, which rectifies,

$$h^1_{\text{NL}} = WA^2\big[\cos q_{\text{AFM}}\tau - \cos q_{\text{AFM}}(2t+\tau)\big]\theta(t), \quad (2)$$

a step switched on at the probe that carries the delay phase but no t dependence. Fed into the E_{12} oscillator it gives $M_{\text{NL}} \propto \cos(q_{\text{AFM}}\tau)[1 - \cos E_{12}t]$: a factorised, rank-one peak—an E1-dark magnon made visible by borrowing the Tm dipole. The energy for the frequency mismatch is drawn from the pulse edge, an intra-pulse difference-frequency process: stretching the pulse from $\sigma = 0.25$ to 1.0 at fixed amplitude collapses the peak by 4.5 decades, while switching off the direct Tm drive changes it by $< 10^{-4}$. Term ablation isolates the pathway completely: the peak dies with the Zeeman leg (0.02) and with W_1^{yy} (0.03), and ignores everything else. *Signature*: $\propto A_{\text{Fe}}^2$, temperature-flat below ~ 10 K, CEF linewidth along ω_t.

The magnon, seen through the ion: $(q_{\text{AFM}}, q_{\text{AFM}}) = 0.28$. With $F_z \equiv 0$ the magnon diagonal in this channel cannot be iron emission. It is the Tm dipole, forced at the magnon frequency by the energy face—$S_xS_z\lambda^1$ contributes $G_z\,\delta S_x\,\lambda^1$, linear in the magnon fluctuation, and $S_y^2\lambda^1$ the pump-magnon×probe-magnon cross term; the ablation runs apportion the peak between the two components (0.43 without the first, 0.64 without the second), the signature of one tensor sharing a task. A driven oscillator responds at the driving frequency, not its own, and the line itself certifies it: the sideband sits at 0.8999 THz—the magnon, not the

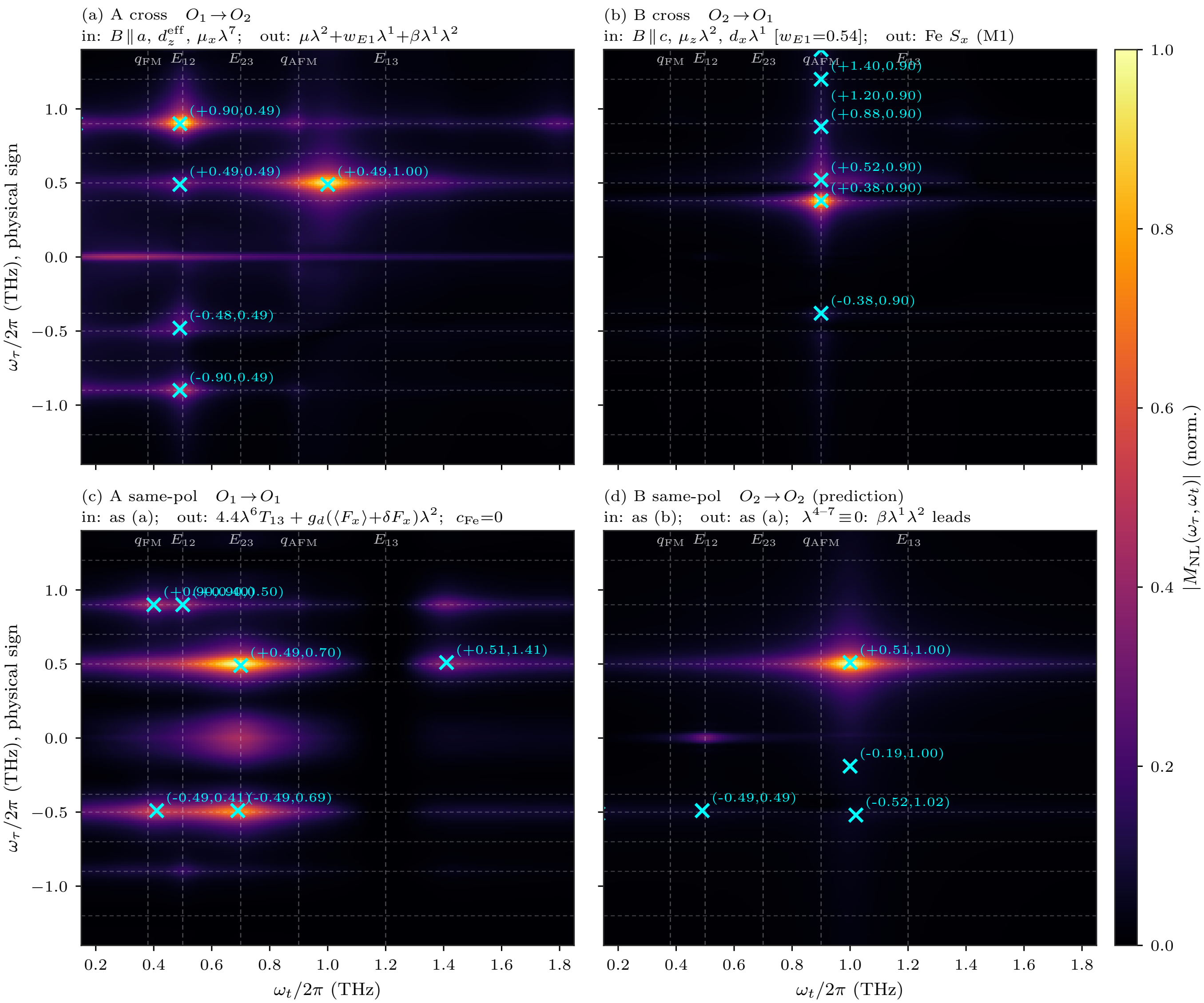

FIG. 1. The atlas: four polarisation channels from one Hamiltonian at one operating point. Dashed guides mark the five mode energies; crosses are blind-census maxima; linear amplitude colour scale; the $\omega_\tau = 0$ pump–probe row is retained in the geometry-A panels (auto-scaled to the measured relative amplitude) and excluded from the census.

$q_{\rm FM}+E_{12} = 0.876$ combination—and the transfer coefficient $|\lambda^2|_{0.90}/|m_x|_{0.90} = 0.14$–$0.16$ is drive-independent over a $20\times$ range, a forced linear response evaluated off resonance.

B. The energy face, read backward

Geometry B, cross-polarised: $(E_{12}, q_{\rm AFM}) = 0.42$. Here the roles invert: a CEF coherence is stored, and a magnon radiates. $E \,\|\, a$ owns the $1 \leftrightarrow 2$ block, so the pulse's *electric* field writes the E_{12} coherence through $d_x\lambda^1$; the energy face hands the stored coherence to the quasi-antiferromagnetic magnon, which radiates through the Fe m_x dipole. No field enters the conversion: the polarisation ownership of d_x performs the geometry selection that a field-assisted vertex would otherwise be invoked for—this drive component does not exist in geometry A, which is why the backward conversion never contaminates the forward peak. Three facts pin the assignment. The *quadrature* matters: writing the same coherence magnetically converts far less cleanly, flooding the map with drive-squared diagonals, whereas the electric writing at $W = 0.01$ adds the transfer row at the *unshifted* line position with strays below 0.17. The interference of the two drive quadratures places the row: with $\mathrm{sign}(d_x\mu_z) < 0$ it sits at $\omega_\tau = 0.52$, exactly where it is measured—the relative sign is thereby a prediction for the crystal-field wavefunctions. And the electric component's amplitude is linear in w_{E1}; the measured 0.43–0.46 returns $w_{E1} = 0.54$. Term ablation resolves the internal anatomy: removing the d_x writing halves the transfer, removing $S_xS_z\lambda^1$ costs a quarter, and the remainder flows through the magnetically written quadrature and the S_y^2 mixing route—every route ultimately rings the real q_{AFM} mode, and the emission stays pinned at 0.900 in all of them. *Signature*: falls as the population difference $n_1 - n_2$ ($0.46 \to 0.36$ from 0 to 10 K), the sharpest thermal discriminator against the temperature-flat magnon–magnon peak below; and, riding the order parameter through the static vertex ($\propto G_z$), the transfer must collapse at the spin reorientation.

The Fe sector alone: $(q_{\mathrm{FM}}, q_{\mathrm{AFM}}) = 1.00$. $H \parallel c$ launches the quasi-ferromagnetic magnon; single-ion anisotropy and the Dzyaloshinskii–Moriya interaction couple the branches, and the stored q_{FM} phase is re-emitted on q_{AFM}. No Fe–Tm coupling enters at any vertex, which is why this peak is temperature-flat: the control experiment that the conversion peaks are measured against.

C. The moment face, linear and bilinear

Geometry A, same-polarised. The analysed moment is m_x, and symmetry admits three contributions: the CEF d_z line (λ^6), the order-induced dipole $g_d(\langle F_x\rangle + \delta F_x)\lambda^2$, and the Fe F_x itself. Table II gives the channel against the digitised measurement, with the cross channels held fixed.

The linear readout: $(q_{\mathrm{AFM}}, E_{12}) = 0.30$. The magnon occupies the delay axis and the ion radiates at E_{12} into the same-polarised channel—possible only through d_z^{eff}, and required by reciprocity, because the same d_z^{eff} is what lets $E \parallel c$ create the E_{12} coherence in the first place. The digitised map contains it at 0.30; the model places it at 0.39, fed jointly by the condensed readout and the low-frequency side of the tones below. This peak is the direct measurement of $g_d\langle F_x\rangle$.

TABLE II. Geometry-A same-polarised channel: measured amplitudes against the model (overlap-gated delay apodisation; self-absorption of the $E1$ emission at the E_{13} line, optical depth $\simeq 6$ on both the emission column and the delay-axis FID, $\simeq 1$ at E_{12}). The μ_{13}^z *and* μ_{13}^x weights fit to zero (fully dark $1 \to 3$).

peak	exp.	model	pathway
$(E_{12}, 0)$	1.00	—	rectified storage (see text)
(E_{12}, E_{23})	1.00	1.00	coherence→pop.→ ρ_{23}
$(E_{12}, 0.40)$	0.45	0.40	cascade tone $2(E_{23}-E_{12})$, adjacent to q_{FM}
$(E_{12}, 0.90)$	0.40	0.43	cascade tone $2E_{23}-E_{12}$, degenerate with q_{AFM}
$(0, E_{12})$	0.40	—	pump–probe row
$(q_{\mathrm{AFM}}, E_{12})$	0.30	0.39	magnon→CEF via d_z^{eff}
$(q_{\mathrm{AFM}}, q_{\mathrm{AFM}})$	absent	0.13	magnon diagonal; absence bounds c_{Fe}
(E_{13}, E_{23})	0	0.00	2Q row, killed by FID reabsorption
$(E_{12}, 1.15)$	0.15	0.02	E_{13} wing; bounds $d_z(1 \to 3) \leq 0.006$
$(q_{\mathrm{AFM}}, 1.41)$	0.30	0.31	SFG tone $q_{\mathrm{AFM}}+E_{12}$
$(q_{\mathrm{AFM}}, 0.40)$	obs.	0.43	DFG tone $q_{\mathrm{AFM}}-E_{12}$
$(E_{12}, 1.41)$	obs.	0.36	E_{12}-delay partner of the SFG
$(-E_{12}, E_{23})$	< 0.15	0.7	**open** (residual R1, point-ion floor)

The bilinear readout: the SFG/DFG triplet $(q_{\mathrm{AFM}}, 1.41)$*,* $(q_{\mathrm{AFM}}, 0.40)$*,* $(E_{12}, 1.41)$. The fluctuating part of the moment face is a product emitter—the stored q_{AFM} magnon beating against the E_{12} coherence on one dipole—and radiates at $q_{\mathrm{AFM}} \pm E_{12} = 1.40/0.40$ THz with the magnon on the delay axis, plus an E_{12}-delay partner at 1.40. All three tones are present in the measured map. This channel is structurally unique in the entire interaction space: of the roughly forty vertices, emission moments and quadratic products the taxonomy admits, it is the only one whose 2D map is *led* by the magnon-delay row, because both of its factors are written at first order by the same pulse—the stored magnon pays no conversion penalty. Term ablation confirms exactly this: the tones survive removal of both W components (0.95–1.02) and die only with the Zeeman or electric drive legs. One number is fitted (the condensation ratio $s = 0.42$), after which the sum tone lands at the measured amplitude, the difference tone merges into the measured $(q_{\mathrm{AFM}}, E_{12})$ blob, and the partner appears at 0.36. The sum tone escapes one absorption linewidth *outside* the opaque E_{13} window (transmission 0.68 at 1.39 against 0.002 at line centre), which is why it is visible while the true E_{13} line centre is dark. Together with the on-site β these tones complete the $\chi^{(2)}$-like triad of the inversion-free rare-earth site, *in emission*: SHG ($\beta\lambda^1\lambda^2$, degenerate), SFG and DFG ($g_d\,\delta F_x\lambda^2$, non-degenerate, cross-subsystem).

The SHG line: geometry A cross, $(E_{12}, 2E_{12}) = 0.96$. The pump stores E_{12}; the detection axis is its second harmonic at $2E_{12} = 1.0000$ THz. A moment linear in the qutrit coordinates cannot radiate a harmonic; the quadratic term does, cleanly—one peak, 5.8× its own diagonal. The assignment is surgical under ablation: the harmonic survives removal of the Zeeman drive and of both W components at 1.00–1.03 and dies to 0.005 without the electric E_{12} writing. *Discriminators*: the line sits at exactly $2E_{12}$ and *tracks* E_{12}, so it does not soften at the spin reorientation as any magnon-derived feature would; its width is twice the E_{12} linewidth; and the same moment must double the magnon-forced component of the dipole, predicting a $(q_{\mathrm{AFM}}, 2q_{\mathrm{AFM}})$ line at 1.79 THz.

D. The bare qutrit, and what is absent

The dominant transfer: $(E_{12}, E_{23}) = 1.00$*, via a population.* With the single-action route closed by the dark $1 \to 3$ dipole (below), the surviving pathway acts twice: one probe interaction converts the stored E_{12} coherence into a level-2 population that still carries the $\omega_\tau = E_{12}$ phase, and a second converts that population into the emitting E_{23} coherence. Co-rotating, it lands non-rephasing—the observed side—and involves μ_{13} at no vertex, which is why it survives while its E_{13} partner is throttled.

Cascade tones at the magnon frequencies. The same driven three-level cascade radiates intrinsic tones at $2(E_{23}-E_{12}) = 0.40$ and $2E_{23}-E_{12} = 0.90$ THz on the λ^6 line—numerically coincident with $q_{\mathrm{FM}} = 0.38$ and $q_{\mathrm{AFM}} = 0.90$, but pure qutrit dynamics: in the simulation they persist at 0.89–0.97 with the Fe drive removed entirely, and die without the $2 \leftrightarrow 3$ drive leg (0.05–0.08), confirming the double-ρ_{23} pathway. The level scheme of $TmFeO_3$ is genuinely conspiratorial here—three separate near-degeneracies between cascade combinations and magnon frequencies—and only term-resolved calculation separates the two readings. The experiment can too: the low tone sits at 0.40, more than a linewidth above $q_{\mathrm{FM}} = 0.38$, and a cascade tone survives the spin reorientation with CEF width, while a magnon sideband would soften and sharpen.

The rectified line: $(E_{12}, 0) = 1.00$*, the strongest measured feature.* The same double action, taken at zero output frequency, rectifies the stored coherence into a slow magnetisation; analysed through $S_{\mathrm{DC}}(\tau)$ it appears as a pair of sharp lines at ± 0.49 THz, symmetric in delay sign as a population grating must be (1.00/0.85 measured). Its *magnitude* carries separate physics: under coherent pumping alone the rectified moment is orders of magnitude below the resonant peaks, so the measured dominance of this line is the optical readout of the *incoherent* population channel—absorbed pulse energy returning through the CEF populations to a quasistatic change of the ordered moment, the same reservoir that the measured post-pulse growth of the geometry-B signal demands and that no coherent pathway in this space can supply (every coherent channel inherits its source's $T_2 \lesssim 8$ ps). Both observables belong to the population face $SS\lambda^3$, cleanly separated from the coherent atlas.

Absences as measurements. The measured map contains delay frequencies $\omega_\tau \in \{E_{12}, q_{\mathrm{AFM}}\}$ only. A bright μ_{13}^z would store an E_{13} coherence at first order and produce an (E_{13}, E_{23}) peak with exactly the dipole product of the observed (E_{12}, E_{23}): the absence of the whole row is a selection rule. The data force more: the $(E \parallel c, H \parallel a)$ analyser physically contains $\mu_x \lambda^5$, and at the point-charge value $\mu_{13}^x = 2.39$ that single term would exceed the entire measured map by two orders of magnitude—μ_{13}^x fits to zero. The $1 \to 3$ transition is magnetically dark in *every* polarisation; its large measured oscillator strength ($f = 8.7$) is electric, and precisely that strong $E1$ line makes the sample opaque at 1.20 THz. The two measured E_{13}-emission features sit at 1.15 and 1.25 THz—one absorption linewidth below and above the line—the textbook signature of self-reversal, and the same reabsorption acting on the delay-axis FID removes the two-quantum (E_{13}, E_{23}) row *exactly* (model $0.33 \to 0.00$ with the filter). Finally, the magnon diagonal $(q_{\mathrm{AFM}}, q_{\mathrm{AFM}})$ is the one feature of the same-polarised channel that genuinely requires iron emission—and the measured map does not contain it. That null bounds the Fe:Tm emission efficiency from above and completes a picture that is uniform across the atlas: the iron sublattice is loud in geometry B, where the magnon owns the analysed moment and is driven resonantly, and quiet in geometry A, where everything measured arrives through the rare-earth ion.

Geometry B, same-polarised: a one-line prediction. This channel shares its detection operator with geometry-A cross, so it is a prediction with no freedom left. The linear readouts are suppressed in M_{NL}; the channel collapses onto its quadratic term, and the prediction is a *single* broad feature, the second harmonic $(E_{12}, 2E_{12}) = 1.00$, everything else below 0.07. Since β is fixed by the geometry-A parity, the channel is a zero-parameter test: a map led by anything else falsifies the quadratic moment, and an observed rectified pair at $(\pm E_{12}, \omega_t \to 0)$ would falsify the exclusion of population readout.

TABLE III. Term-by-term verification (single-seed ablation runs; ratios of feature amplitudes to the full model).

removed	what dies (ratio)	verdict
Fe Zeeman leg	A-cross anchor 0.02; tones 0.03	required
d_z^{eff} drive	anchors, SHG 0.005; tones 0.1	required
$\mu_x \lambda^7$ drive	anchor 0.02; cascade tones 0.05	required
$W\, S_y^2 \lambda^1$	A-cross anchor 0.03	required
$W\, S_x S_z \lambda^1$	sideband 0.64; B transfer 0.76	required (shared
β (detect)	SHG $0.95 \to 0.01$	required
λ^6 (detect)	same-pol anchor 0.06	required
$\delta F_x \lambda^2$ (detect)	SFG/DFG triplet 0.02–0.05	required
T_{13} (column / row)	0.67 line-centre peak / 2Q row 0.32	required
d_x drive (geom. B)	transfer 0.49	dominant leg
$BS\lambda$ family	nothing (bitwise)	inert, exact
$SS\lambda^3$ + thermal	coherent features $\leq 5\%$	incoherent only
λ^4, F_z (detect)	only the 1.15 wing / nothing	bounded / spect

V. TERM-BY-TERM ACCOUNTING

Table III closes the loop: every component of the final Hamiltonian is certified necessary by a named measured feature that dies without it, and everything else that symmetry allows is zero, bounded, or inert. The minimal content of the theory is two cross-subsystem constants (W and g_d), one on-site anharmonicity (β), one measured drive ratio (w_{E1}), and one propagation filter. The interaction between the crystal-field ladder and the spin waves of $TmFeO_3$ is, in the audited sense, completely characterised by Eq. (1).

VI. IMPLICATIONS FOR RARE-EARTH MAGNETISM

Three statements generalise beyond $TmFeO_3$.

The rare-earth ion is a coherent transducer, not a passive thermal bath. In the standard picture of the orthoferrites the R-ion enters through its free energy: thermally occupied CEF levels shift the Fe anisotropy, and reorientation follows. The atlas shows the same coupling surface operating *coherently*: the energy face converts quanta between magnon and CEF coherence within picoseconds, in both directions—the geometry-A anchor forward, the geometry-B transfer backward—phase-resolved and mode-resolved. The couplings that reorient the statics and the vertices that convert the dynamics are one object read at two levels of description, which is presumably why terahertz control of the Fe order through the rare-earth channel is possible at all [7, 8]. The sharpest consequence is collective: every dressed feature—the transfer peaks, the $(q_{\mathrm{AFM}}, E_{12})$ readout, the SFG/DFG triplet—rides $\langle F_x \rangle$ or G_z and must collapse *together* at the reorientation, while the bare features (the magnon–magnon peak, the SHG line, the cascade tones) survive it. A temperature scan through T_{SR} cleaves the atlas into its two halves along exactly the line Eq. (1) draws.

Kramers parity controls the atlas. The cleanliness of $\mathrm{TmFeO_3}$—sharp unshifted lines, perturbative assignable vertices, a readable atlas—is the singlet protection at work: first-order exchange is symmetry-blocked, so nothing hybridises strongly and everything converts weakly. In a Kramers member of the family (Er, Yb) no such protection exists: linear exchange coupling is allowed, the R modes and magnons form mixed magnon–CEF excitations, and the corresponding 2D atlas should be qualitatively different—dominated by hybridisation shifts and mixed-mode linewidths rather than by weak, assignable conversion vertices. The non-Kramers orthoferrites (Tm, Pr, Tb) are the clean laboratories; comparing a Kramers and a non-Kramers member under the same protocol would elevate this to a family-wide organising principle.

Order-induced moments carry mandatory fluctuation sidebands. If a transition moment exists only because an order parameter condensed ($d_z^{\mathrm{eff}} \propto \langle F_x \rangle$), then the same coupling constant necessarily radiates sum- and difference-frequency tones with the order parameter's own fluctuations, at amplitudes locked to the condensed line by a single ratio. This is a universal signature, not a $\mathrm{TmFeO_3}$ accident: any order-induced line in any magnetoelectric—multiferroic electromagnons, exchange-induced dipoles, field-induced moments—must carry these sidebands, and 2DCS is the instrument that separates them from cascades, because the delay axis distinguishes a stored order-parameter fluctuation from a stored coherence. Here the sidebands were not decoration: they account for the highest-frequency cross peak in the spectrum.

VII. PREDICTIONS AND TESTS

The structure constrains itself as follows. Any peak emitting at q_{AFM} through the iron dipole inherits the $15\times$ sharper magnon linewidth; $(q_{\mathrm{FM}}, q_{\mathrm{AFM}})$ is temperature-flat while $(E_{12}, q_{\mathrm{AFM}})$ falls as $n_1 - n_2$; the transfer row sits at the 1D absorption frequency exactly (a static mixer would place it below, at any coupling strength); $\mathrm{sign}(d_x \mu_z) < 0$, checkable against the crystal-field wavefunctions; rotating either polarisation away from c must revive the $\omega_\tau = E_{13}$ rows; the $2E_{12}$ line tracks E_{12} and does not soften at the reorientation, whereas the entire dressed sector must vanish there; the geometry-B same-polarised map must be led by the $2E_{12}$ harmonic essentially alone; and adiabatically long pulses collapse the geometry-A cross peak. Propagation adds fingerprints: a lineout across $\omega_t = 1.1$–1.3 at $\omega_\tau = q_{\mathrm{AFM}}$ must show a self-reversal *doublet* around a dark centre at 1.20; the geometry-B cross channel must carry a d_z^{eff} doublet at $\omega_t = 0.45/0.54$ at the ~ 0.17 level; and every wing amplitude scales with thickness, so a thin sample revives the E_{13} line centre and the two-quantum row together. The same-polarised E_{12}-row tones must sit at 0.40 and 0.90 with CEF width and survive the reorientation—at the low tone a full linewidth above q_{FM}, a direct positional test—and any recovered $(q_{\mathrm{AFM}}, q_{\mathrm{AFM}})$ diagonal would raise the c_{Fe} bound and must appear with the magnon width. The SFG/DFG triplet is over-constrained: locked ratios, position at $q_{\mathrm{AFM}}+E_{12}$ exactly (at 1.20 it would be a quadratic moment instead, and the dark E_{13} window separates the cases), tracking of the softening magnon, and collective collapse at the reorientation. The temperature scan through T_{SR} remains the decisive experiment: it cleaves the atlas into dressed and bare halves and measures g_d by the size of the collapse.

One residual survives, named and characterised. *R1, the rephasing remnant* $(-E_{12}, E_{23})$: the point-ion qutrit has an exact floor of 0.47–0.58 of the dominant transfer (verified against a standalone three-level integrator; no coupling in the admitted space and no drive quadrature or sign removes it), while the measured negative-delay side carries only the rectified and q_{FM} features—and the SFG/DFG tones contribute rephasing partners of their own there (model 0.5–0.7), sharpening the same question. This residual, the self-reversal wing amplitudes, and the geometry-B d_z^{eff} doublet strength funnel into a single measurable object: the sample's polarisation-resolved optical depths at the three CEF lines. One transmission measurement closes it or breaks the propagation layer cleanly.

VIII. CONCLUSION

The interaction between crystal-field excitations and spin waves in $\mathrm{TmFeO_3}$ is intrinsically nonlinear: time reversal protects the singlet ion from its magnetic environment at first order, and what survives is a second-order dressing with exactly two faces—an energy face $W\,SS\,\lambda^1$ that converts and never radiates, and a moment face $g_d\,E_z(F_x\lambda^2)$ that drives and radiates, linearly through its condensed part and bilinearly through its fluctuations, as internal magnon–CEF sum- and difference-frequency generation. Two-dimensional terahertz spectroscopy resolves this algebra term by term: two constants, one on-site anharmonicity, one measured drive ratio and one self-absorption filter reproduce the polarisation-resolved atlas from measured inputs, every assignment verified by switching off individual terms, and the absences doing quantitative work—a fully dark $1 \to 3$ dipole, a bounded iron emission weight, an excluded quadratic-moment sector. Because the energy face is the spin-reorientation coupling surface promoted from thermodynamics to operators, the atlas is a coherent, mode-resolved measurement of the mechanism that moves the statics of the orthoferrite family, and its sharpest prediction is collective: at the reorientation the dressed half of the atlas dies together while the bare half survives. The program—bound the interaction space by symmetry, assign every feature and every absence, verify every term, and let the atlas overdetermine the theory—applies to any magnet whose 2D map spans distinct subsystems, and turns 2DCS from a spectroscopy of modes into a spectroscopy of Hamiltonian terms.

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Derivations, numerical protocol and the complete falsification ledger are in the Supplemental Material; the long-form note and all run recipes accompany the data.

[1] J. Lu, X. Li, H. Y. Hwang, B. K. Ofori-Okai, T. Kurihara, T. Suemoto, and K. A. Nelson, Phys. Rev. Lett. **118**, 207204 (2017).

[2] S. Mukamel, *Principles of Nonlinear Optical Spectroscopy* (Oxford University Press, New York, 1995).

[3] T. Yamaguchi, J. Phys. Chem. Solids **35**, 479 (1974).

[4] A. V. Kimel, A. Kirilyuk, A. Tsvetkov, R. V. Pisarev, and Th. Rasing, Nature **429**, 850 (2004).

[5] *Resolving the spin reorientation and crystal-field transitions in $TmFeO_3$ with terahertz transient*, Sci. Rep. **6**, 23648 (2016).

[6] *Terahertz magnon-polaritons in $TmFeO_3$*, ACS Photonics **5**, 1375 (2018).

[7] S. Baierl *et al.*, *Nonlinear spin control by terahertz-driven anisotropy fields*, Nat. Photonics **10**, 715 (2016).

[8] S. Schlauderer *et al.*, *Temporal and spectral fingerprints of ultrafast all-coherent spin switching*, Nature **569**, 383 (2019).